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Stochastic Processes

Lecture 14

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The Stopped Process

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Let $\mathbf{X} = \{X_n\}_{n\in\mathbb{N}}$ be a random process.
Let $\mathscr{G} = \{\mathscr{G}_n : n \in \mathbb{N}\}$ be the natural filtration adapted to the process $\mathbf{X}$.
Let τ be a stopping time adapted to $\mathscr{G}$. Define

$$\mathbf{Y}_\tau := (X_1, \ldots, X_\tau), \qquad \mathbf{Y}_{\tau(\omega)}(\omega) = (X_1(\omega), \ldots, X_{\tau(\omega)}(\omega)), \quad \omega \in \Omega.$$

Then, $\mathbf{Y}_\tau$ is called the **stopped process**.

Important Question

Two important questions are of relevance.

- What is the codomain of $\mathbf{Y}_\tau$?
- If $\mathbb{R}^\star$ is the co-domain, what is $\mathscr{B}(\mathbb{R}^\star)$?

The Stopped Process

- The length of $\mathbf{Y}_\tau$ is not fixed
- Define $\mathbf{Z}_\tau$ as

$$\mathbf{Z}_\tau \;:=\; (\tau, \mathbf{Y}_\tau) \;=\; (\tau, X_1, \ldots, X_\tau).$$

  Notice that $\mathbf{Y}_\tau$ and $\mathbf{Z}_\tau$ have one-one correspondence
- Then, the codomain of $\mathbf{Z}_\tau$ is

$$\mathbb{R}^\star \;=\; \bigcup_{n\in\mathbb{N}} \Big(\{n\} \times \mathbb{R}^n\Big).$$

- Each element of $\mathbb{R}^*$ is of the form $(n, x_1, \ldots, x_n)$ for some $n \in \mathbb{N}$ and $x_1, \ldots, x_n \in \mathbb{R}^n$
- The Borel σ-algebra on $\mathbb{R}^\star$ is given by

$$\mathscr{B}(\mathbb{R}^\star) \;=\; \Big\{A \subseteq \mathbb{R}^\star : \quad \underbrace{A \cap \Big(\{n\} \times \mathbb{R}^n\Big)}_{\text{slice of dimension } n} \in \mathscr{B}(\{n\} \times \mathbb{R}^n) \quad \forall\, n \in \mathbb{N}\Big\}.$$

- The generating class for $\mathscr{B}(\mathbb{R}^\star)$ is given by

$$\mathscr{E} \;=\; \Big\{\{n\} \times B : \quad n \in \mathbb{N},\;\; B \in \mathscr{B}(\mathbb{R}^n)\Big\}.$$

The Stopped Process

- We now prove that $\mathbf{Z}_\tau : \Omega \to \mathbb{R}^\star$ is indeed a random variable
- To prove that $\mathbf{Z}_\tau$ is a random variable, it suffices to show that for any $B^\star \in \mathscr{E}^\star$,

$$\mathbf{Z}_\tau^{-1}(B^\star) \in \mathscr{F}.$$

- Taking $B^\star = \{n\} \times B$ for fixed $n \in \mathbb{N}$ and $B \in \mathscr{B}(\mathbb{R}^n)$, we have

$$\mathbf{Z}_\tau^{-1}(\{n\} \times B) \;=\; \{\tau = n\} \cap \{(X_1, \ldots, X_n) \in B\} \;\in\; \mathscr{G}_n \;\subseteq\; \mathscr{F}.$$

What is $\sigma(\mathbf{Z}_\tau)$?

The Stopped σ-Algebra

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.

Definition (Stopped σ-Algebra)

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ be a random process.
Let $\mathscr{G} = \{\mathscr{G}_n : n \in \mathbb{N}\}$ be the natural filtration adapted to the process $\mathbf{X}$.
Let τ be a stopping time adapted to $\mathscr{G}$. Then,

$$\mathscr{G}_\tau \;:=\; \Big\{ E \in \mathscr{F} : \quad E \cap \{\tau = t\} \in \mathscr{G}_t \quad \forall\ t \Big\}$$

is called the **stopped σ-algebra**.

Remark:

- **Exercise:** $\mathscr{G}_\tau$ is a σ-algebra of subsets of Ω
- $\sigma(\mathbf{Z}_\tau) = \mathscr{G}_\tau$ (will be given as a homework question)
- If $\mathcal{T} = \mathbb{R}$ or $\mathbb{R}_+$, then

$$\{\tau = t\} \;\leftarrow\; \{\tau \leq t\}$$

  in the definition of the stopped σ-algebra

Strong Independence Property

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.

Lemma (Strong Independence Property)

Let $\mathbf{X} = \{X_n\}_{n\in\mathbb{N}}$ *be an* **independent process** *with respect to* $\mathscr{F}$.
Let τ *be a* **stopping time** *adapted to the natural filtration of the process* $\mathbf{X}$. *Then, for each* $n \in \mathbb{N}$,

$$(\tau, X_1, \ldots, X_\tau) \perp\!\!\!\perp X_{\tau+n} \;\Big|\; \tau.$$

That is, conditioned on the stopping time, the future is independent of the past and the present.

Remark:

- For any (deterministic) $n \in \mathbb{N}$, we know

$$(X_1, \ldots, X_n) \perp\!\!\!\perp (X_{n+1}, X_{n+2}, \ldots).$$

- Strong independence property says that the above is true if n is replaced by a stopping time τ

What if τ is Not a Stopping Time?

- Let $X_1, X_2, \ldots \overset{\text{i.i.d.}}{\sim} \mathrm{Ber}(0.5)$, and let

$$\tau \;:=\; \inf\{n \geq 1 : \; X_n = X_{n+1} = 1\}.$$

- Clearly, $X_\tau = X_{\tau+1} = 1$, and therefore

$$(\tau, X_1, \ldots, X_\tau) \;\not\perp\!\!\!\perp\; X_{\tau+1}.$$

Proof of Lemma 1

- To prove the result, we will prove that for any $x \in \mathbb{R}$,

$$\{X_{\tau+n} \leq x\} \perp\!\!\!\perp A \,\Big|\, \tau \qquad\qquad \forall\, A \in \mathscr{G}_\tau.$$

  That is, we have to show that

$$\mathbb{P}\Big(A \cap \Big\{X_{\tau+n} \leq x\Big\}\Big|\tau\Big) \;=\; \mathbb{P}\Big(A\Big|\tau\Big) \cdot \mathbb{P}\Big(\Big\{X_{\tau+n} \leq x\Big\}\Big|\tau\Big).$$

- For each $m \in \mathbb{N}$, we have

$$\begin{aligned}
\mathbb{P}\Big(A \cap \{X_{\tau+n} \leq x\}\Big|\{\tau = m\}\Big) &= \frac{\mathbb{P}\Big(A \cap \{\tau = m\} \cap \{X_{\tau+n} \leq x\}\Big)}{\mathbb{P}(\{\tau = m\})} \\
&= \frac{\mathbb{P}\Big(A \cap \{\tau = m\}\Big) \cdot \mathbb{P}\Big(\{X_{m+n} \leq x\}\Big)}{\mathbb{P}(\{\tau = m\})} \\
&= \mathbb{P}\Big(A\Big|\{\tau = m\}\Big) \cdot \mathbb{P}\Big(\{X_{m+n} \leq x\}\Big) \\
&= \mathbb{P}\Big(A\Big|\{\tau = m\}\Big) \cdot \mathbb{P}\Big(\{X_{\tau+n} \leq x\}\Big|\{\tau = m\}\Big).
\end{aligned}$$

Markov Chains

Markov Chain

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.

Definition (Markov Chain)

A process $\mathbf{X} = \{X_t : t \in \mathcal{T}\}$ is called a **Markov chain** if for any $t \in \mathcal{T}$,

$$\{X_s : s < t\} \perp\!\!\!\perp \{X_s : s > t\} \mid X_t.$$

That is, **conditioned on the present value, the future is independent of the past**.

Markov property in terms of CDFs:

For any $m, n \in \mathbb{N}$, times

$$s_1 < \cdots < s_m < t < t_1 < \cdots < t_n,$$

and values $(x_1, \ldots, x_m) \in \mathbb{R}^m$, $(y_1, \ldots, y_n) \in \mathbb{R}^n$, and $x \in \mathbb{R}$,

$$\mathbb{P}\Big(\Big\{\underbrace{X_{s_1} \le x_1, \ldots, X_{s_m} \le x_m}_{\text{before } t}, \underbrace{X_{t_1} \le y_1, \ldots, X_{t_n} \le y_n}_{\text{after } t}\Big\} \Big| \{X_t \le x\}\Big)$$

$$= \mathbb{P}\Big(\Big\{X_{s_1} \le x_1, \ldots, X_{s_m} \le x_m \mid X_t \le x\Big\}\Big) \cdot \mathbb{P}\Big(\Big\{X_{t_1} \le y_1, \ldots, X_{t_n} \le y_n \mid X_t \le x\Big\}\Big).$$

Markov Chain

- If $\mathcal{T} = \mathbb{N}$, we refer to the Markov chain as a **discrete time Markov chain (DTMC)**
- When talking about Markov chains, the range of the random variables plays an important role
- If the range is a finite set, the Markov chain is called a **finite-state** Markov chain
- If range is a countable set, the Markov chain is called a **countable-state (or denumerable)** Markov chain
- The range is itself called the **state space**

DTMC on a Finite State Space

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$.

Definition (Markov Chain)

A process $\mathbf{X} = \{X_n : n \in \mathbb{N}\}$ is called a **DTMC on a finite state space $\mathcal{X}$** if for any $m, n \in \mathbb{N}$, times

$$s_1 < \cdots < s_m < t < t_1 < \cdots < t_n,$$

and values $(x_1, \ldots, x_m) \in \mathcal{X}^m$, $(y_1, \ldots, y_n) \in \mathcal{X}^n$, and $x \in \mathcal{X}$,

$$\mathbb{P}\Big(\Big\{\underbrace{X_{s_1} = x_1, \ldots, X_{s_m} = x_m}_{\text{before } t}, \underbrace{X_{t_1} = y_1, \ldots, X_{t_n} = y_n}_{\text{after } t}\Big\} \Big| \Big\{X_t = x\Big\}\Big)$$

$$= \mathbb{P}\Big(\Big\{X_{s_1} = x_1, \ldots, X_{s_m} = x_m\Big\} \Big| \Big\{X_t = x\Big\}\Big) \cdot \mathbb{P}\Big(\Big\{X_{t_1} = y_1, \ldots, X_{t_n} = y_n\Big\} \Big| \Big\{X_t = x\Big\}\Big).$$

Consequences of Markov Property

For a DTMC $\mathbf{X} = \{X_n\}_{n\in\mathbb{N}}$ on a finite state space $\mathcal{X}$:

- For all $m, n \in \mathbb{N}$, $(x_1, \ldots, x_m) \in \mathcal{X}^m$, $(y_1, \ldots, y_n) \in \mathcal{X}^n$, and $x \in \mathcal{X}$,

$$\mathbb{P}\Big(\underbrace{\Big\{X_{t_1} = y_1, \ldots, X_{t_n} = y_n\Big\}}_{\text{after } t}\Big|\Big\{X_t = x, \underbrace{X_{s_1} = x_1, \ldots, X_{s_m} = x_m}_{\text{before } t}\Big\}\Big)$$

$$= \mathbb{P}\Big(\Big\{X_{t_1} = y_1, \ldots, X_{t_n} = y_n\Big\}\Big|\Big\{X_t = x\Big\}\Big).$$

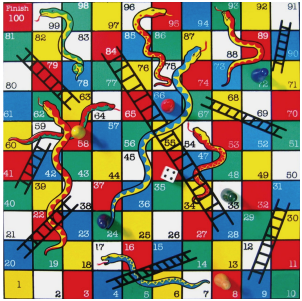
- For any $n \in \mathbb{N}$, $(x_1, \ldots, x_n) \in \mathcal{X}^n$, and $x \in \mathcal{X}$,

$$\mathbb{P}\Big(\Big\{X_{n+1} = x\Big\}\Big|\Big\{X_1 = x_1, \ldots, X_n = x_n\Big\}\Big) = \mathbb{P}\Big(\Big\{X_{n+1} = x\Big\}\Big|\Big\{X_n = x_n\Big\}\Big).$$

- **FDDs:** For any $n \in \mathbb{N}$, $t_1 < t_2 < \cdots < t_n$, and $(x_1, \ldots, x_n) \in \mathcal{X}^n$,

$$\mathbb{P}\Big(\Big\{X_1 = x_1, \ldots, X_n = x_n\Big\}\Big) = \mathbb{P}(\{X_1 = x_1\}) \cdot \prod_{t=2}^{n} \mathbb{P}\Big(\Big\{X_t = x_t\Big\}\Big|\Big\{X_{t-1} = x_{t-1}\Big\}\Big).$$

Example



- $\mathcal{X} = \{1, \ldots, 100\}$
- Suppose $X_n = 6$ at some time n
- Then, $X_{n+1} \in \{7, 8, 31, 10, 11, 12\}$
- For all $x, x_1, \ldots, x_{n-1} \in \mathcal{X}$, we have

$$\mathbb{P}\Big(\Big\{X_{n+1} = x\Big\}\Big|\Big\{X_n = 6, \underbrace{X_{n-1} = x_{n-1}, \ldots, X_1 = x_1}_{\text{trajectory before } n}\Big\}\Big)$$

$$= \mathbb{P}\Big(\Big\{X_{n+1} = x\Big\}\Big|\Big\{X_n = 6\Big\}\Big).$$

Example: Random Walk

Let $\{X_n\}_{n\in\mathbb{N}}$ be an $\mathbb{N}$-valued i.i.d. process. For each $n \in \mathbb{N}$, let $S_n = \sum_{i=1}^{n} X_i$.

- For any $n \in \mathbb{N}$ and $s_1 < s_2 < \cdots < s_n < s_{n+1}$,

$$\begin{aligned}
\mathbb{P}\Big(\Big\{S_{n+1} = s_{n+1}\Big\}\Big|\Big\{S_1 = s_1, \ldots, S_n = s_n\Big\}\Big) &= \mathbb{P}\Big(\Big\{X_{n+1} = s_{n+1} - s_n\Big\}\Big|\Big\{S_1 = s_1, \ldots, S_n = s_n\Big\}\Big) \\
&= \mathbb{P}\Big(\Big\{X_{n+1} = s_{n+1} - s_n\Big\}\Big) \\
&= \mathbb{P}\Big(\Big\{S_{n+1} = s_{n+1}\Big\}\Big|\Big\{S_n = s_n\Big\}\Big)
\end{aligned}$$

- Thus, the process $\{S_n\}_{n\in\mathbb{N}}$ is a DTMC with state space $\mathcal{S} = \mathbb{N}$

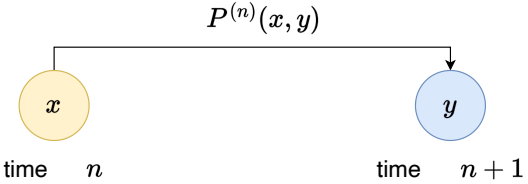
Transition Probability Matrix (TPM)

Transition Probability Matrix

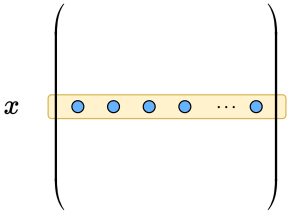
Definition (Transition Probability Matrix)

Let $\{X_n\}_{n\in\mathbb{N}}$ be a DTMC on a finite state space $\mathcal{X}$.
The **transition probability matrix (TPM)** of the Markov chain at any fixed time $n \in \mathbb{N}$ is a matrix $P^{(n)} = [P^{(n)}(x,y)]_{x,y\in\mathcal{X}}$ defined as

$$P^{(n)}(x,y) = \mathbb{P}\Big(\Big\{X_{n+1} = y\Big\}\Big|\Big\{X_n = x\Big\}\Big), \qquad x, y \in \mathcal{X}.$$



Transition Probability Matrix



- For each $x \in \mathcal{X}$, $\quad \sum_{y \in \mathcal{X}} P^{(n)}(x, y) = 1.$
- A matrix with non-negative entries and row sums equal to 1 is called a **row stochastic matrix**
- $P^{(n)}$ is row stochastic for every $n \in \mathbb{N}$
- Each row of $P^{(n)} \quad \equiv \quad$ PMF on $\mathcal{X}$
- For instance, if $\mathcal{X} \subseteq \mathbb{R}$, then

$$\sum_{y \in \mathcal{X}} y^2 \, P^{(n)}(x, y) = \mathbb{E}\left[X_{n+1}^2 \middle| \left\{X_n = x\right\}\right],$$

  where $\mathbb{E}$ above is $P^{(n)}(x, \cdot)$
- $P^{(n)}$ has a right eigenvector with eigenvalue 1
- $P^{(n)} \cdot \mathbf{1} = \mathbf{1}$, where $\mathbf{1}$ is the all-ones vector

Time Homogeneous DTMC

Definition (Time Homogeneous DTMC)

A DTMC $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ on a finite state space $\mathcal{X}$ and having TPMs $\{P^{(n)}\}_{n \in \mathbb{N}}$ is called **time homogeneous** if

$$P^{(n)} = P^{(n+1)} \qquad \forall n \in \mathbb{N}.$$

That is, $P^{(n)}$ is same for all n.
In this case, we simply write P to denote the common TPM.

Example: Random Walk

Let $X_1, X_2, \ldots$ be i.i.d. on $\{-1, +1\}$ with PMF

$$\mathbb{P}(\{X_1 = 1\}) \ = p \ = 1 - \mathbb{P}(\{X_1 = -1\}).$$

Let $S_0 := 0$, and for each $n \in \mathbb{N}$, let $S_n = \sum_{i=1}^{n} X_i$.
What is the TPM of $\{S_n\}_{n \in \mathbb{N}}$?